CO1 - Assessment tool - 1

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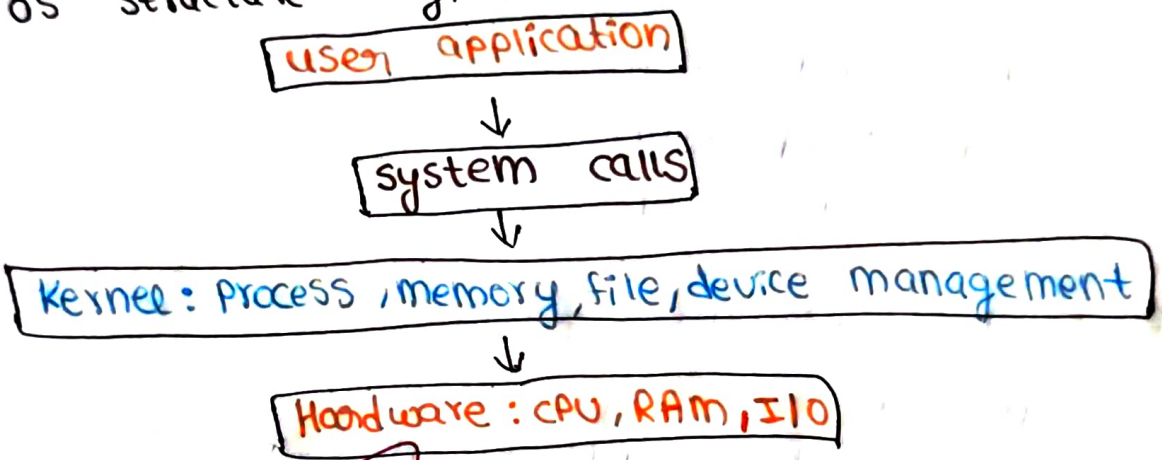
1) operating system for multiple users and processes:

a) Types of operating system:

* multi-user OS: It supports multiple users.
* multitasking OS: Runs multiple tasks.
* multiprocessing OS: OS uses multiple CPUs.
* suitable OS: Linux or UNIX.

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b) OS structure diagram:



c) Process system calls:

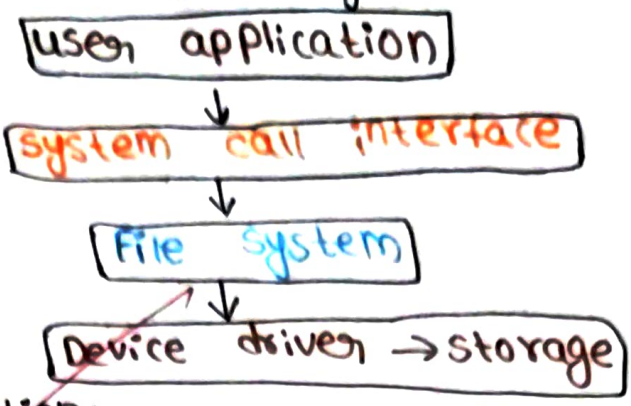
* Fork() - creates a process.
* exec() - Executes a program.
* wait() - waits for a process.
* exit() - Terminates a process.

2) File and device management:

a) categories of system calls:

* File management: open(), read(), write(), close().
* Device management: ioctl(), read(), write().
* process management: fork(), exit().

b) Layered structure diagram:



c) File operation:

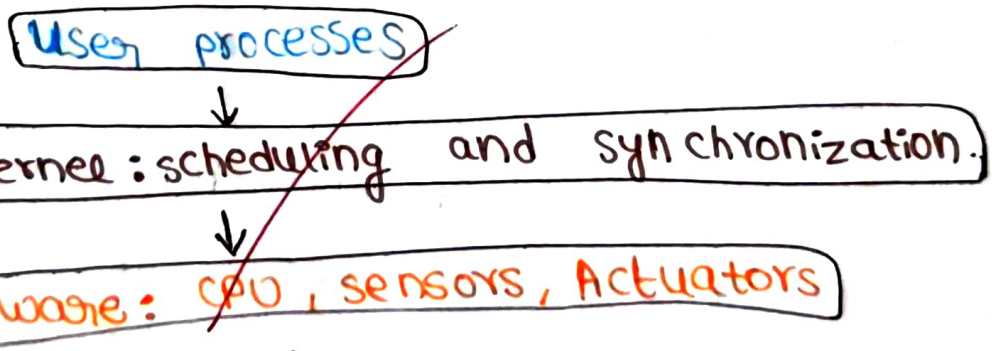
* open(): opens a file.
* read(): Read data.
* close(): closes the file.

3) Real-time operating system:

a) Type and suitability:

* RTOS responds within a specified time.
* Hard RTOS is used for critical industrial machinery.
* It provides fast, predictable responses.
* Examples: QNX and VxWorks.

b) Interaction diagram:



c) Synchronization functions:

* sem_wait(): waits for a semaphore.

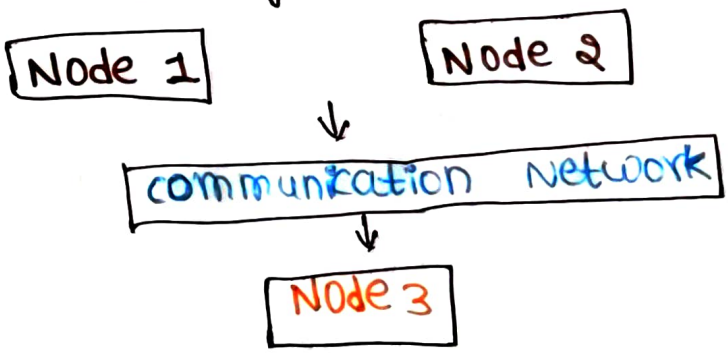
* sem_post(): Releases a semaphore.
* pthread_mutex_lock(): Locks a resource.
* pthread_mutex_unlock(): unlocks a resource.

4) Distributed operating system:

a) classification and comparison:

* Distributed OS connects multiple computers through a network.
* Distributed OS shares resources among nodes.
* centralized OS manages resources mainly through one central system.

b) Architecture diagram:



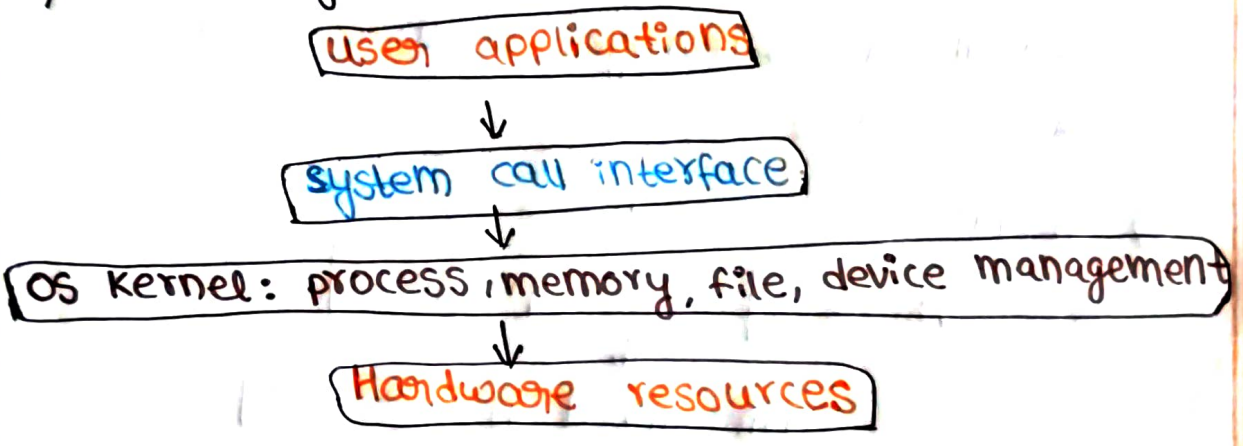
c) IPC system calls:

* pipe() - creates a communication channel.
* fork() - creates a process.
* write() - sends data.
* read() - Receives data.
* close() - closes the channel.

5) a) OS Resource management:

* process management - controls processes.
* CPU scheduling - Allocates CPU time.
* memory management - Allocates RAM.
* Device management - controls I/O devices.
* File management - stores and retrieves files.

b) Block diagram:



c) System call examples:

* mmap(): maps memory.
* munmap(): Removes a memory mapping.
* open(): opens a device.
* Read(): Reads device data.
* write(): writes device data.
* ioctl(): controls device specific operations.